Kai Wang 王凯

Postdoctoral Research Associate
ICC & CEA, Durham University

wkcosmology@gmail.com
www.KosmosWalker.com
0000-0002-3775-0484

Experience

Postdoctoral Research Associate 2024 – present
ICC & CEA, Durham University

KIAA Fellow 2022 – 2024
Kavli Institute for Astronomy and Astrophysics, Peking University

Education

Ph.D. in Astronomy 2017 – 2022
Tsinghua University

Visiting Scholar 2019 – 2021
University of Massachusetts Amherst

B.S. in Astronomy 2013 – 2017
University of Science and Technology of China (USTC)

Publication Summary

48	**20**	**530+**	**15+**
publications	first/corresponding author	citations	h-index
36 since 2023	16 since 2023	420+ since 2023	full list on NASA/ADS

Selected Publications

1. **Inflow-driven galaxy evolution – I. Revealing the physics of the fundamental metallicity relation**
K. Wang, C. Baugh, N. F. Boardman, S. Bose, Z. Cai, S. Cole, C. S. Frenk, C. Lacey, P. Norberg, Y. Peng, I. Santos-Santos, A. Saintonge, T. Theuns, E. Wang
Submitted to MNRAS (2026), arXiv:2608.04784
I derived a unified framework for the chemical evolution of galaxies from the two mass-continuity equations, providing the first self-consistent explanation of the fundamental metallicity relation (FMR) and its redshift invariance to $z \sim 3$. The FMR is a projection of the gaseous FMR: its parameterisation encodes how star formation efficiency varies with stellar mass and redshift, and its shape shows that galaxies evolve from inflow-driven to equilibrium.

2. **Inflow-driven galaxy evolution – II. A hierarchy in the scaling relations of star-forming galaxies**
K. Wang, C. Baugh, S. Bose, S. Cole, C. S. Frenk, C. Lacey, J. F. Navarro, K. A. Oman, A. Saintonge, T. Theuns
Submitted to MNRAS (2026), arXiv:2610.12394
I identified a hierarchy among the three primary scaling relations of star-forming galaxies: the star-forming main sequence (SFMS), the gas-to-stellar mass relation and the mass–metallicity relation (MZR). Halo formation time sets the SFMS; combined with the star formation efficiency, the SFMS gives the gas-to-stellar mass relation; and combined with the mass-loading factor, the gas-to-stellar mass relation gives the MZR. This hierarchy also explains three higher-dimensional relations: the FMR, its gaseous counterpart (gFMR) and the fundamental plane of star formation.

3. **Forged in Quenching: morphological transformation across star-forming and quiescent galaxies in EAGLE**
K. Wang, C. Baugh, S. Bose, S. Cole, C. S. Frenk, H. Fu, C. Lacey, S. Lu, A. Ludlow, P. Norberg, Y. Peng, K. L. Proctor, I. Santos-Santos, F. Shankar, T. Theuns, E. Wang, T. Wang, V. Wild
Submitted to MNRAS (2026), arXiv:2609.16187
Using EAGLE, I showed that morphology actively drives SMBH growth and the quenching of central galaxies, rather than being passively inherited through progenitor bias. I also showed that merger-driven SMBH growth in semi-analytic models cannot reproduce the size difference between star-forming and quiescent galaxies, pointing to the need for a secular, morphology-dependent SMBH growth channel.

4. **An efficient and robust method to estimate halo concentration based on the method of moments**
K. Wang, H. Mo, Y. Chen, J. Schaye
MNRAS, 527, 10760 (2023), arXiv:2310.00200
I proposed a robust moment-based estimator of halo concentration, now adopted in the simulation post-processing tools SOAP and AHF and in the FLAMINGO and COLIBRE cosmological simulations.

5. **Dissecting the two-halo galactic conformity effect: the dependence of star formation activity on the large-scale environment for central galaxies**
K. Wang, Y. Peng, Y. Chen
MNRAS, 523, 1268 (2023), arXiv:2304.06886
I found that backsplash galaxies explain the dependence of central galaxy quenching on large-scale environment and give rise to two-halo galactic conformity in both IllustrisTNG and SDSS.

Grants and Awards

The 2026 Physics Awards for Excellence Universe Makers Team 2026
The 2025 Physics Awards for Excellence NAM 2025 Local Organising Committee 2025
T. D. Lee Postdoctoral Fellowship declined 2024
KIAA Postdoctoral Fellow start-up research funding CNY 50,000 2022
KIAA Postdoctoral fellowship CNY 504,000 2022
MUST fellowship declined 2022
Comprehensive scholarship (2nd class) of Tsinghua University CNY 5,000 2020
China Scholarship Council award for visiting scholars US$45,600 2019
AMD scholarship (2nd class) of Tsinghua University CNY 5,000 2018
Future Scholar Scholarship of Tsinghua University CNY 150,000 2017
The annual scholarship of National Astronomical Observatories, CAS CNY 3,000 2017
National Inspirational Award CNY 5,000 2016
National Astronomy Training Base CNY 20,000 2016
National Astronomy Training Base CNY 20,000 2015

Supervision and Mentorship

Oscar Hickman, graduate, Durham University 2026 – present
Less is More: unbiased galaxy clustering from N-body merger trees
Sienna Folkes-Miller, Level-4, Durham University 2026 – present
Tracing the Invisible: how galaxies reveal the dark Universe
Thomas Power, Level-4, Durham University 2025 – 2026
The Dance of the Milky Way and Andromeda: a celestial ballet across cosmologies
Thomas Richardson, Level-4, Durham University 2025 – 2026
Where Shadows Follow Light: baryon-induced halo contraction in hydrodynamical simulations

Haochen Jiang, undergraduate, University of Science and Technology of China 2023 – 2025
Dissecting the quenching of massive central galaxies in TNG arXiv:2510.24886

Xunda Sun, graduate, University of Chinese Academy of Sciences 2023 – 2024
Characterising the spatial distribution of the metal content of galaxies in FIRE-2 arXiv:2409.09290

Zeyu Gao, graduate, Peking University 2022 – 2024
Decoding the SEDs of galaxies with a prior from hydrodynamical simulations arXiv:2408.07749

Chengyu Ma, graduate, University of Science and Technology of China 2023 – 2024
Revisiting the fundamental metallicity relation with observations and simulations arXiv:2407.21716

Zhijun Zhang, Bachelor's thesis, Peking University 2022 – 2023
Identifying protoclusters in high-redshift photometric surveys

Teaching

Level-1 Physics Tutor 2024 – 2025
Durham University
Year-long weekly tutorials for five undergraduates covering mechanics, electromagnetism and quantum mechanics, taught interactively with students working at the board. Peer-observed by the Director of Education, Prof. Michael Hunt: "a well planned and conducted tutorial", noting the practice of "going beyond the model answer to explain underlying concepts".

Tutorial Assistant, ExGal Summer School on Astrophysical Simulations 2025
Durham University

Guest Lecturer, Cosmology and Galaxy Evolution 2023
Peking University

Teaching Assistant, Observational Cosmology 2017
Tsinghua University

Teaching Assistant, Particle Cosmology 2017
University of Science and Technology of China

Teaching Assistant, General Relativity 2016
University of Science and Technology of China

Outreach

Summer Science Exhibition, The Universe Makers team 2026
The Royal Society, London
The Summer Science Exhibition is the Royal Society's flagship annual public engagement festival celebrating the cutting edge of UK research and innovation. Welcoming over 10,000 total visitors—including 2,000 secondary school students and teachers—the high-capacity, six-day event features 13 interactive flagship exhibits and over 60 specialised talks and panels delivered by 100+ world-leading speakers.

Undergraduate Open Day, Demonstrator 2026
Durham University
The Undergraduate Open Day is Durham University's flagship two-day recruitment event for prospective applicants and parents, showcasing its top-5 UK standing. Spanning 16 undergraduate colleges and dozens of departments, the event utilises a massive staff and student ambassador network to deliver presentations, exhibitions, and campus tours.

Schools' Science Festival, Demonstrator 2025
Durham University
The Durham University Schools' Science Festival is an annual outreach programme designed to inspire Year 9 and Year 10 students. The 2025 festival welcomed over 700 pupils from 22 schools across the region to participate in interactive, hands-on scientific projects.

Celebrate Science, Demonstrator 2024
Palace Green, Durham
Celebrate Science is Durham University's annual public festival aimed at children aged 7–11, families, and school groups, designed to make complex research accessible. The event attracts between 5,000 and 7,500+ attendees annually, featuring over 200 volunteers to deliver interactive STEM workshops.

Service

Referee for MNRAS (since 2024), ApJ (since 2023) and A&A (since 2022)

Organiser, CosmICConnections 2026 – present
Durham University
CosmICConnections is a hybrid seminar and discussion series held every four weeks, providing an informal forum for local and visiting researchers to present their work, exchange ideas, and foster collaborations across astronomy and cosmology.

Organiser, Friday Lunchtime Astronomy Talks (FLAT) 2025 – 2026
Durham University
The Friday Lunchtime Astronomy Talk (FLAT) is a weekly seminar series featuring around 60 talks annually by local and visiting early-career researchers, fostering scientific discussion and collaboration among students and postdoctoral researchers.

LOC member, Looking into the Dark: Frenk@75 2026
The Royal Society, London
The Looking into the Dark: Frenk@75 conference, held at the Royal Society in London, celebrated the 75th birthday of Professor Sir Carlos S. Frenk CBE FRS, recognising his contributions to cosmology and galaxy formation and his lasting influence on the students, postdoctoral researchers, and collaborators he has worked with throughout his career.

SOC member and panellist, Galaxy Chemistries Across Cosmic Time 2026
National Astronomy Meeting (NAM) 2026

SOC member, The Golden Era of Gravitational Lensing 2025
National Astronomy Meeting (NAM) 2025

LOC member, National Astronomy Meeting (NAM) 2025 2025
Durham University
NAM 2025, hosted by Durham University, brought together over 1,400 attendees across scientific and public events, featured more than 500 talks, and strengthened collaboration across the UK astronomy community while showcasing Durham as an international hub for astronomy research.

Co-lead, morning arXiv journal club, Durham University 2025

Postdoc representative, Faculty Candidate Interview Committee, KIAA 2023, 2024

Organiser, weekly Galaxy Party, KIAA 2023

Organiser, Postdoc Science Day, KIAA 2022

Organiser, Speaker Lunch, Tsinghua Center for Astrophysics 2018 – 2019

Talks

Recordings and slides: www.KosmosWalker.com/presentation

Invited seminars. University of California, Santa Cruz (2023, 2024, 2026); Kavli IPMU, University of Tokyo (2021, 2026); National Astronomical Observatories, CAS (2026); Peking University (2023, 2025, 2026); Tsinghua University (2022, 2024, 2025, 2026); Nanjing University (2026); Westlake University (2026); Liverpool John Moores University (2025); University of St Andrews (2025); University of Science and Technology of China (2025); Shanghai Jiao Tong University (2025); South-Western Institute for Astronomy Research, Yunnan University (2024); Huazhong University of Science and Technology (2024); Carnegie Observatories (2024); KIPAC, Stanford University (2024).

Conference talks. DEX-XXII, Edinburgh (2026); SWIFTCON, Leiden (2025); L-GALAXIES Workshop, Berlin (2025); National Astronomy Meeting, Durham (2025); Expanding the Boundaries of Dark Matter Haloes, Shanghai (2025); Mock Barcelona, Barcelona (2024); The Co-evolution of Galactic Ecosystems and their Large-scale Environments, Hangzhou (2024); 2nd Shanghai Assembly on Cosmology and Structure Formation, Shanghai (2023); Collaboration Workshop on Cosmology and Galaxy Formation, Shanghai (2023); 25th CAS Guoshoujing Symposium on Galaxies and Cosmology, Huangshan (2023); Conference on Star Formation and Nuclear Activity in Galaxies, Nanjing (2023); 11th and 10th Prime Focus Spectrograph Collaboration Meetings, Pasadena (2019) and Shanghai (2018).

Publications

First- and corresponding-author papers ([†] Corresponding author)

1. **K. Wang**[†], C. Baugh, S. Bose, S. Cole, C. S. Frenk, C. Lacey, J. F. Navarro, K. A. Oman, A. Saintonge, T. Theuns, *Inflow-driven galaxy evolution – II: A hierarchy in the scaling relations of star-forming galaxies*, Submitted to MNRAS (2026), arXiv:2610.12394
2. **K. Wang**[†], C. Baugh, N. F. Boardman, S. Bose, Z. Cai, S. Cole, C. S. Frenk, C. Lacey, P. Norberg, Y. Peng, I. Santos-Santos, A. Saintonge, T. Theuns, E. Wang, *Inflow-driven galaxy evolution – I. Revealing the physics of the fundamental metallicity relation*, Submitted to MNRAS (2026), arXiv:2608.04784
3. **K. Wang**[†], C. Baugh, S. Bose, S. Cole, C. S. Frenk, H. Fu, C. Lacey, S. Lu, A. Ludlow, P. Norberg, Y. Peng, K. L. Proctor, I. Santos-Santos, F. Shankar, T. Theuns, E. Wang, T. Wang, V. Wild, *Forged in Quenching: Morphological Transformation across Star-forming and Quiescent Galaxies in EAGLE*, Submitted to MNRAS (2026), arXiv:2609.16187
4. **K. Wang**[†], C. Baugh, S. Bose, S. Cole, C. S. Frenk, C. Lacey, S. Lu, P. Norberg, I. Santos-Santos, F. Shankar, T. Theuns, T. Wang, *Ruffled Feathers: Merger-driven galaxy size growth and structural transformation in EAGLE*, Submitted (2026), arXiv:2608.05835
5. **K. Wang**[†], P. Norberg, A. Fattahi, L. E. Strigari, *Local Group analogues in a cosmological context – I. Relating velocity structure to the cosmic web*, Submitted (2026), arXiv:2601.08419
6. **K. Wang**[†], J. Schaye, A. Benítez-Llambay, E. Chaikin, C. S. Frenk, F. Huško, R. J. McGibbon, S. Ploeckinger, A. J. Richings, M. Schaller, J. W. Trayford, *Gravitational potential drives the concentration dependence of the stellar mass–halo mass relation*, MNRAS, 546, stag110 (2026), arXiv:2511.20165
7. **K. Wang**[†], *The origin of the galaxy size–stellar metallicity relation – I. A semi-analytical perspective*, MNRAS, 545, staf2113 (2026), arXiv:2510.02573
8. H. Jiang, E. Wang[†], **K. Wang**[†], C. Ma, X. Kong[†], *Dissecting the Mass Quenching in TNG50: Galaxy Size Determines the Quenching Mode*, ApJ, 994, 220 (2025), arXiv:2510.24886
9. **K. Wang**[†], Y. Peng[†], *Testing Galaxy Formation Models with the Stellar Mass–Halo Mass Relations for Star-forming and Quiescent Galaxies*, ApJ, 980, 233 (2025), arXiv:2408.07743
10. C. Ma, **K. Wang**[†], E. Wang[†], Y. Peng, H. Jiang, H. Yu, C. Jia, Z. Chen, H. Li, X. Kong, *Revisiting the Fundamental Metallicity Relation with Observation and Simulation*, ApJL, 971, L14 (2024), arXiv:2407.21716

11. **K. Wang**†, H. J. Mo, Y. Chen, H. Wang, X. Yang, J. Wang, Y. Peng, Z. Cai, *Characterizing the assembly of dark matter haloes with protohalo size histories: I. Redshift evolution, relation to descendant haloes, and halo assembly bias*, MNRAS, 528, 2046 (2024), arXiv:2309.01039
12. **K. Wang**†, H. J. Mo, Y. Chen, J. Schaye, *An efficient and robust method to estimate halo concentration based on the method of moments*, MNRAS, 527, 10760 (2024), arXiv:2310.00200
13. **K. Wang**†, X. Wang†, Y. Chen, *Environmental Dependence of the Mass-Metallicity Relation in Cosmological Hydrodynamical Simulations*, ApJ, 951, 66 (2023), arXiv:2305.08161
14. **K. Wang**†, Y. Chen, Q. Li, X. Yang, *Late-formed haloes prefer to host quiescent central galaxies – I. Observational results*, MNRAS, 522, 3188 (2023), arXiv:2304.07189
15. **K. Wang**†, Y. Peng†, Y. Chen, *Dissect two-halo galactic conformity effect for central galaxies: the dependence of star formation activities on the large-scale environment*, MNRAS, 523, 1268 (2023), arXiv:2304.06886
16. **K. Wang**†, H. J. Mo, C. Li, Y. Chen, *Relating galaxies across different redshift to study galaxy evolution*, MNRAS, 520, 1774 (2023), arXiv:2211.00485
17. **K. Wang**†, H. J. Mo, C. Li, Y. Chen, *Finding protoclusters to trace galaxy evolution – I. The finder and its performance*, MNRAS, 505, 3892 (2021), arXiv:2104.12223
18. **K. Wang**†, H. J. Mo, C. Li, J. Meng, Y. Chen, *Identifying galaxy groups at high redshift from incomplete spectroscopic data – I. The group finder and application to zCOSMOS*, MNRAS, 499, 89 (2020), arXiv:2006.05426
19. **K. Wang**, L. Santos, J.-Q. Xia, W. Zhao†, *Thermal gravitational-wave background in the general pre-inflationary scenario*, JCAP, 01, 053 (2017), arXiv:1608.04189
20. Y.-F. Wang, **K. Wang**†, W. Zhao, *Smoothing methods comparison for CMB E- and B-mode separation*, RAA, 16, 59 (2016), arXiv:1511.01220

Co-author papers

21. H. Fu et al., *Mergers may not be the main channel for galaxy stellar mass growth at redshift $z > 5$: predictions from cosmological hierarchical models and simulations*, Submitted (2026), arXiv:2610.17575
22. Y.-C. Wang et al., *Massive Galaxy Halos Contain Less Inner Dark Matter Than Predicted*, Submitted (2026), arXiv:2609.19132
23. C. Ma et al., *Fountain pattern of baryon cycle revealed in galaxy ecosystems*, Submitted (2026), arXiv:2609.21374
24. K. Zhao et al., *Reconstructing the Projected Dark Matter Field across 0.1–100 Mpc Scales from the SDSS Survey*, ApJL, accepted (2026), arXiv:2609.19295
25. D. Zhao et al., *No Detectable One-halo Galactic Conformity Signal with Halo-mass Estimates Consistent with Weak-lensing Constraints*, ApJL, 1009, L7 (2026), arXiv:2609.11313
26. N. F. Boardman et al., *Beyond the fundamental metallicity relation: galaxy sizes encode the link between inflow and metallicity*, MNRAS, 550, stag1263 (2026), arXiv:2606.13124
27. W. Kong et al., *Cooler Phases of the Circumgalactic Medium Are More Centrally Concentrated: Constraints from Multiphase Absorption Lines*, ApJ, 1006, 49 (2026), arXiv:2606.04793
28. T. Yang et al., *A Census of Quiescent Galaxies across $0.5 < z < 8$ with JWST/MIRI: Mass-dependent Number Density Evolution of Quiescent Galaxies in the Early Universe*, ApJL, 1000, L42 (2026), arXiv:2510.12235
29. H. Sun et al., *The Bigfoot: A Footprint of a Coma Cluster Progenitor at $z = 3.98$*, ApJL, 993, L49 (2025), arXiv:2508.21356
30. X. Sun et al., *The Physical Origin of Positive Metallicity Radial Gradients in High-redshift Galaxies: Insights from the FIRE-2 Cosmological Hydrodynamic Simulations*, ApJ, 986, 179 (2025), arXiv:2409.09290

31. C. Jia et al., *Potential-driven Metal Cycling: JADES Census of Gas-phase Metallicity for Galaxies at $1 < z < 7$*, ApJL, 986, L24 (2025), arXiv:2504.18820
32. D. Zhao et al., *From Halos to Galaxies. VI. Improved Halo Mass Estimation for SDSS Groups and Measurement of the Halo Mass Function*, ApJ, 979, 42 (2025), arXiv:2408.12442
33. Z. Gao et al., *From Halos to Galaxies. X. Decoding Galaxy SEDs with Physical Priors and Accurate Star Formation History Reconstruction*, ApJ, 979, 66 (2025), arXiv:2408.07749
34. C. Lyu et al., *From Halos to Galaxies. IX. Estimate of Halo Assembly History for SDSS Galaxy Groups*, ApJ, 972, 108 (2024), arXiv:2407.03409
35. T. Wang et al., *Black holes regulate cool gas accretion in massive galaxies*, Nature, 632, 1009 (2024), arXiv:2311.07653
36. Q. Li et al., *Black Hole–Halo Mass Relation from UNIONS Weak Lensing*, ApJL, 969, L25 (2024), arXiv:2402.10740
37. J. Meng et al., *Measuring Galaxy Abundance and Clustering at High Redshift from Incomplete Spectroscopic Data: Tests on Mock Catalogs*, ApJ, 964, 161 (2024), arXiv:2008.13733
38. C. Lyu et al., *From Halos to Galaxies. VII. The Connections between Stellar Mass Growth History, Quenching History, and Halo Assembly History for Central Galaxies*, ApJ, 959, 5 (2023), arXiv:2310.10733
39. Y. Chen et al., *Massive dark matter haloes at high redshift: implications for observations in the JWST era*, MNRAS, 526, 2542 (2023), arXiv:2304.13890
40. Y. Chen et al., *A conditional abundance matching method of extending simulated halo merger trees to resolve low-mass progenitors and subhalos*, MNRAS, 525, 1254 (2023), arXiv:2301.08972
41. Q. Li et al., *Groups and Protocluster Candidates in the CLAUDS and HSC-SSP Joint Deep Surveys*, ApJ, 933, 9 (2022), arXiv:2205.05517
42. Y. Chen et al., *MAHGIC: a Model Adapter for the Halo-Galaxy Inter-Connection*, MNRAS, 507, 2510 (2021), arXiv:2106.03984
43. Z. Wang et al., *The clustering of galaxies in the DESI imaging legacy surveys DR8: I. The luminosity and color dependent intrinsic clustering*, Sci. China Phys. Mech. Astron., 64, 289811 (2021), arXiv:2106.14159
44. Y. Chen et al., *How to empirically model star formation in dark matter haloes – I. Inferences about central galaxies from numerical simulations*, MNRAS, 504, 4865 (2021), arXiv:2009.12467
45. Y. Chen et al., *Relating the Structure of Dark Matter Halos to Their Assembly and Environment*, ApJ, 899, 81 (2020), arXiv:2003.05137
46. J. Ye et al., *Superconducting cosmic strings as sources of cosmological fast radio bursts*, Eur. Phys. J. C, 77, 720 (2017), arXiv:1705.10956
47. L. Santos et al., *Statistical imprints of CMB B-type polarization leakage in an incomplete sky survey analysis*, JCAP, 01, 043 (2017), arXiv:1612.03564
48. L. Santos et al., *Probing the statistical properties of CMB B-mode polarization through Minkowski functionals*, JCAP, 07, 029 (2016), arXiv:1510.07779